

“I saw in a dream a table where all the elements fell into place as required.”

Dmitri Mendeleev

Charles Darwin
See pages
134–135

1871

Human evolution

Charles Darwin published *The Descent of Man*, a follow-up volume to *On the Origin of Species*. In his new work, he discussed the way in which humans had evolved and his theory of sexual selection. Although Darwin had hesitated before publishing the book, it did not create as much controversy as his first.

1872

Scientific exploration

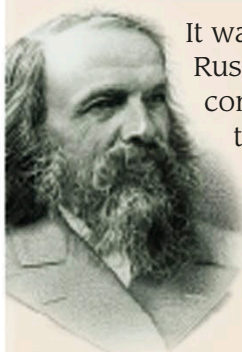
HMS *Challenger*, a British ship, set sail on a four-year voyage to explore the world's oceans. It carried a team of scientists, whose many discoveries helped lay the foundations of oceanography (the branch of science that involves the study of seas and oceans).



The zoology laboratory on board HMS *Challenger*

1875

1834–1907 DMITRI MENDELEEV

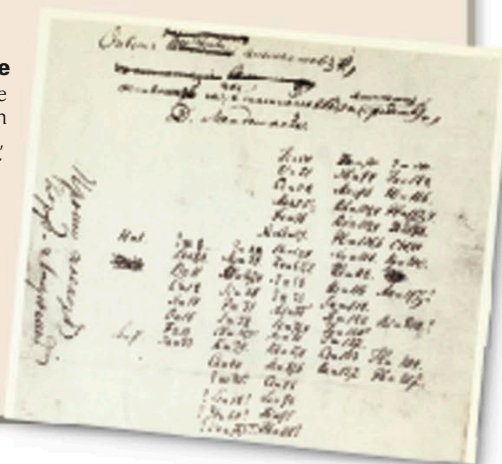


It was as a Professor of Chemistry in St. Petersburg, Russia, that Dmitri Mendeleev came up with the concept of a periodic table of elements based on their atomic weight. He later said the idea came to him in a dream while he was preparing a textbook on chemistry. Mendeleev himself called it the Periodic System. The element Mendeleevium, discovered in 1955, was named after him, in honor of his discovery.

The periodic table

Mendeleev realized that if he arranged the 63 known elements in ascending order of atomic weight, he could organize them in groups, or “periods,” sharing similar properties. Using this, he was then able to predict the existence of three more elements—gallium, scandium, and germanium—all of which were found in the next 16 years (see pp.188–189).

Mendeleev's notes

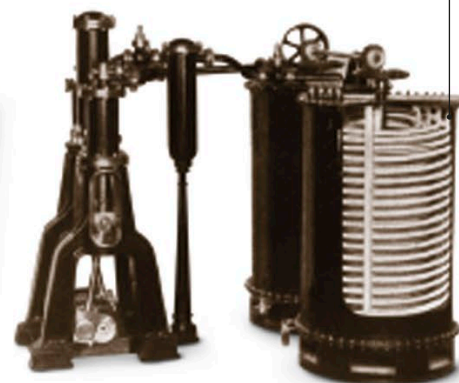


1873

Keeping cool

German engineer Carl von Linde designed the first modern refrigeration system, using liquefied gases, for a brewery in Munich, Germany. Refrigeration rapidly transformed the storage of perishable foodstuffs.

The cooling process in Von Linde's invention isn't too different from what is found in modern refrigerators.



Von Linde's refrigerator